Сетевые технологии

Настройка DHCP, DHCPv6, SLAAC и IPv6 в GNS3

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29 ноября 2025

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Цель работы

Основная цель

Получить навыки настройки DHCP, DHCPv6 Stateless и Stateful, SLAAC и IPv6-адресации в виртуальной среде GNS3.

Выполнение работы

Топология IPv4

Рис. 1: Топология

Настройка VyOS

- Смена имени хоста и доменного имени
- Создание нового пользователя
- Назначение адреса 10.0.0.1/24
- Настройка DHCP-сервера

```
You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name zashikhalieva-gw-01
[edit]
vyos@vyos# set system domain-name zashikhalieva.net
[edit]
vyos@vyos# set system login user zashikhalieva authentication plaintext-passwor
 123456
[edit]
vyos@vyos# delete interfaces ethernet eth0 address dhcp
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
```

DHCP-сервер

- Подсеть: **10.0.0.0/24**
- Диапазон: **10.0.0.2–10.0.0.253**
- DNS: **10.0.0.1**
- Домен: **trseidaliev.net**

```
zashikhalieva@zashikhalieva-gw-01:~$ configure
[edit]
zashikhalieva@zashikhalieva-gw-01# delete system login user vyos
[edit]
zashikhalieva@zashikhalieva-gw-01# commit
[edit]
zashikhalieva@zashikhalieva-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
zashikhalieva@zashikhalieva-gw-01# set interfaces ethernet eth0 address 10.0.0.1
/24
[edit]
zashikhalieva@zashikhalieva-gw-01#
[edit]
zashikhalieva@zashikhalieva-gw-01#
[edit]
zashikhalieva@zashikhalieva-gw-01# set service dhcp-server shared-network-name 1
ashikhalieva domain-name zashikhalieva.net
[edit]
```

Клиент PC1

IP: 10.0.0.2/24

Gateway: 10.0.0.1

DNS: 10.0.0.1

```
ateway IP Address: 0.0.0.0
lient MAC Address: 00:50:79:66:68:00
ption 53: Message Type = Offer
ption 54: DHCP Server = 10.0.0.1
ption 51: Lease Time = 86400
ption 1: Subnet Mask = 255.255.255.0
ption 3: Router = 10.0.0.1
ption 6: DNS Server = 10.0.0.1
ption 15: Domain = zashikhalieva.net

pcode: 1 (REQUEST)
lient IP Address: 10.0.0.2
our IP Address: 0.0.0.0
erver IP Address: 0.0.0.0
ateway IP Address: 0.0.0.0
lient MAC Address: 00:50:79:66:68:00
ption 53: Message Type = Request
ption 54: DHCP Server = 10.0.0.1
ption 50: Requested IP Address = 10.0.0.2
ption 61: Client Identifier = Hardware Type=Ethernet MAC Address = 00:50:79:66:
8:00
ption 12: Host Name = VPCS
```

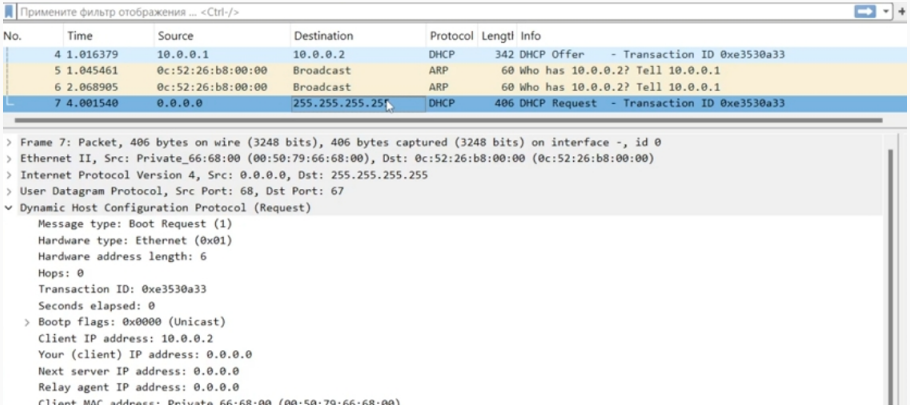
Проверка работы DHCP

```
PCS> show ip

AME         : VPCS[1]
P/MASK      : 10.0.0.2/24
ATEWAY      : 10.0.0.1
NS          : 10.0.0.1
HCP SERVER  : 10.0.0.1
HCP LEASE   : 86391, 86400/43200/75600
OMAIN NAME  : zashikhalieva.net
AC          : 00:50:79:66:68:00
PORT        : 10010
HOST:PORT   : 127.0.0.1:10011
TU          : 1500

PCS> ping 10.0.0.1 -c 2

4 bytes from 10.0.0.1 icmp_seq=1 ttl=64 time=4.043 ms
4 bytes from 10.0.0.1 icmp_seq=2 ttl=64 time=3.377 ms
```

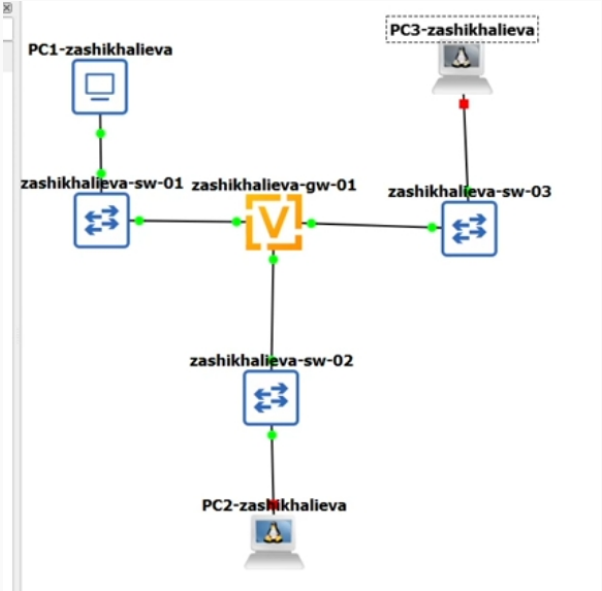
Рис. 5: PC1 show ip

Последовательность обмена

- Discover
- Offer
- Request
- ACK



Новая расширенная топология



Настройка адресов IPv6

- eth1 → 2000::1/64
- eth2 → 2001::1/64

```
[edit]
zashikhalieva@zashikhalieva-gw-01# set interfaces ethernet eth1 address 2000::1
64
[edit]
zashikhalieva@zashikhalieva-gw-01# set interfaces ethernet eth2 address 2001::1
64
[edit]
zashikhalieva@zashikhalieva-gw-01# show interfaces
 ethernet eth0 {
     address 10.0.0.1/24
     hw-id 0c:52:26:b8:00:00
 }
 ethernet eth1 {
+    address 2000::1/64
     hw-id 0c:52:26:b8:00:01
 }
 ethernet eth2 {
+    address 2001::1/64
     hw-id 0c:52:26:b8:00:02
 }
 loopback lo {
 }
[edit]
```


Рис. 8: Interfaces IPv6

Router Advertisement (RA)

- Префикс 2000::/64
- other-config-flag: получение DNS через DHCPv6

```
zashikhalieva@zashikhalieva-gw-01# set service router-advert interface eth1 pre
ix 2000::/64
[edit]
zashikhalieva@zashikhalieva-gw-01# set service router-advert interface eth1 oth
r-config-flag
[edit]
zashikhalieva@zashikhalieva-gw-01#
[edit]
zashikhalieva@zashikhalieva-gw-01#
[edit]
zashikhalieva@zashikhalieva-gw-01# set service dhcpv6-server shared-network-nam
 zashikhalieva-stateless
[edit]
zashikhalieva@zashikhalieva-gw-01# set service dhcpv6-server shared-network-nam
 zashikhalieva-stateless subnet 2000::0/64
[edit]
zashikhalieva@zashikhalieva-gw-01# set service dhcpv6-server shared-network-nam
 zashikhalieva-stateless common-options name-server 2000::1
[edit]
zashikhalieva@zashikhalieva-gw-01# set service dhcpv6-server shared-network-nam
 zashikhalieva-stateless common-options domain-search zashikhalieva.net
[edit]
zashikhalieva@zashikhalieva-gw-01# c
```

DHCPv6 Stateless

Параметры выдаются:

- DNS: 2000::1
- Domain-search: trseidaliev.net

```
                }
            }
        }
    }
    dhcpv6-server {
        shared-network-name zashikhalieva-stateless {
            common-options {
                domain-search zashikhalieva.net
                name-server 2000::1
            }
            subnet 2000::0/64 {
            }
        }
    }
    router-advert {
        interface eth1 {
            other-config-flag
            prefix 2000::/64 {
            }
        }
    }
```

Клиент PC2 (SLAAC)

Адрес: SLAAC (2000::/64)

Маршруты добавлены автоматически

```
udhcpc6: sending select
^C
/ #
/ # ping 2000::1 -c 2
PING 2000::1 (2000::1): 56 data bytes
64 bytes from 2000::1: seq=0 ttl=64 time=2.031 ms
64 bytes from 2000::1: seq=1 ttl=64 time=3.083 ms

--- 2000::1 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 2.031/2.557/3.083 ms
/ # cat /etc/resolv.conf
/ #
```

Рис. 11: PC2 IPv6 SLAAC

DHCPv6 Stateless: получение DNS

```
udhcpc6: sending select
^C
/ #
/ # ping 2000::1 -c 2
PING 2000::1 (2000::1): 56 data bytes
64 bytes from 2000::1: seq=0 ttl=64 time=2.031 ms
64 bytes from 2000::1: seq=1 ttl=64 time=3.083 ms

--- 2000::1 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 2.031/2.557/3.083 ms
/ # cat /etc/resolv.conf
/ #
```

Рис. 12: PC2 dhclient -6

Анализ DHCPv6 Stateless

Примените фильтр отображения ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
22	114.205592	fe80::e52:26ff:feb8…	fe80::42:5aff:fe63:…	ICMPv6	78	Neighbor Advertisement fe80::e52:26ff:feb8:1 (rtr, sol)
23	119.522760	fe80::e52:26ff:feb8…	fe80::42:5aff:fe63:…	ICMPv6	86	Neighbor Solicitation for fe80::42:5aff:fe63:dd00 from 0c:5
24	119.523356	fe80::42:5aff:fe63:…	fe80::e52:26ff:feb8…	ICMPv6	78	Neighbor Advertisement fe80::42:5aff:fe63:dd00 (sol)
25	131.944927	fe80::42:5aff:fe63:…	ff02::1:2	DHCPv6	116	Solicit XID: 0x4c9625 [ROOT-ONLY DOMAIN NAME]CID: 00030001
26	131.953948	fe80::e52:26ff:feb8…	fe80::42:5aff:fe63:…	DHCPv6	162	Advertise XID: 0x4c9625 CID: 0003000102425a63dd00
27	131.954080	fe80::42:5aff:fe63:…	fe80::e52:26ff:feb8…	ICMPv6	210	Destination Unreachable (Port unreachable)
28	131.976412	fe80::42:5aff:fe63:…	ff02::1:2	DHCPv6	134	Request XID: 0x4c9625 [ROOT-ONLY DOMAIN NAME]CID: 00030001
29	131.977177	fe80::e52:26ff:feb8…	fe80::42:5aff:fe63:…	DHCPv6	162	Reply XID: 0x4c9625 CID: 0003000102425a63dd00
30	131.977330	fe80::42:5aff:fe63:…	fe80::e52:26ff:feb8…	ICMPv6	210	Destination Unreachable (Port unreachable)
31	134.998958	fe80::42:5aff:fe63:…	ff02::1:2	DHCPv6	182	Request XID: 0x4c9625 [ROOT-ONLY DOMAIN NAME]CID: 00030001
32	135.000971	fe80::e52:26ff:feb8…	fe80::42:5aff:fe63:…	DHCPv6	162	Reply XID: 0x4c9625 CID: 0003000102425a63dd00

```
> Frame 28: Packet, 134 bytes on wire (1072 bits), 134 bytes captured (1072 bits) on interface -, id 0
> Ethernet II, Src: 02:42:5a:63:dd:00 (02:42:5a:63:dd:00), Dst: IPv6mcast_01:00:02 (33:33:00:01:00:02)
> Internet Protocol Version 6, Src: fe80::42:5aff:fe63:dd00, Dst: ff02::1:2
> User Datagram Protocol, Src Port: 546, Dst Port: 547
v DHCPv6
    Message type: Request (3)
    Transaction ID: 0x4c9625
  > Elapsed time
  > Server Identifier
  > Identity Association for Non-temporary Address
  > Option Request
  v Client Fully Qualified Domain Name
      Option: Client Fully Qualified Domain Name (39)
      Length: 2
    > Flags: 0x00  [CLIENT wants to update its AAAA RRs and SERVER to update its PTR RRs]
    > Root only domain name: ['.' (0)]
  > Client Identifier
```

Включение режима Stateful

- RA: managed-flag
- Диапазон адресов: **2001::100 – 2001::199**
- DNS: 2001::1
- Domain-search: trseidaliev.net

```
[edit]
zashikhalieva@zashikhalieva-gw-01# set service dhcpv6-server shared-network-nam
 zashikhalieva-stateful
[edit]
zashikhalieva@zashikhalieva-gw-01# set service dhcpv6-server shared-network-nam
 zashikhalieva-stateful subnet 2001::0/64
[edit]
zashikhalieva@zashikhalieva-gw-01# set service dhcpv6-server shared-network-nam
 zashikhalieva-stateful subnet 2001::0/64 name-server 2001::1
[edit]
zashikhalieva@zashikhalieva-gw-01# set service dhcpv6-server shared-network-nam
 zashikhalieva-stateful subnet 2001::0/64 domain-search zashikhalieva.net
[edit]
zashikhalieva@zashikhalieva-gw-01# set service dhcpv6-server shared-network-nam
 zashikhalieva-stateful subnet 2001::0/64 address-range start 2001::100 stop 20
1::199
[edit]
zashikhalieva@zashikhalieva-gw-01#
```

PC3 — до получения адреса

```
PC3-zashikhalieva console is now available... Press RETURN to get started.
/ # ip address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
3: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN
 qlen 1000
    link/ether 02:42:b2:11:4f:00 brd ff:ff:ff:ff:ff:ff
    inet6 fe80::42:b2ff:fe11:4f00/64 scope link
       valid_lft forever preferred_lft forever
/ # ip -6 route show
fe80::/64 dev eth0  metric 256
default via fe80::e52:26ff:feb8:2 dev eth0  metric 1024  expires 0sec
/ # cat /etc/resolv.conf
/ # udhcpc6 -i eth0
udhcpc6: started, v1.37.0
udhcpc6: sending discover
udhcpc6: sending select
udhcpc6: IPv6 obtained, lease time 43200
/ #
```

Рис. 15: PC3 SLAAC

PC3 — получение адреса DHCPv6 Stateful

```
   inet 127.0.0.1/8 scope host lo
      valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host
      valid_lft forever preferred_lft forever
: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOW
qlen 1000
   link/ether 02:42:b2:11:4f:00 brd ff:ff:ff:ff:ff:ff
   inet6 fe80::42:b2ff:fe11:4f00/64 scope link
      valid_lft forever preferred_lft forever
 # ip -6 route show
e80::/64 dev eth0  metric 256
efault via fe80::e52:26ff:feb8:2 dev eth0  metric 1024  expires 0sec
 # ping 2001::1 -c 2
ING 2001::1 (2001::1): 56 data bytes
4 bytes from 2001::1: seq=0 ttl=64 time=2.258 ms
4 bytes from 2001::1: seq=1 ttl=64 time=3.630 ms

-- 2001::1 ping statistics ---
 packets transmitted, 2 packets received, 0% packet loss
ound-trip min/avg/max = 2.258/2.944/3.630 ms
 # cat /etc/resolv.conf
earch zashikhalieva.net
ameserver 2001:0000:0000:0000:0000:0000:0000:0001
 #
```


Рис. 16: dhclient Stateful

PC3 — после получения адреса

Полная конфигурация:
- IPv6: 2001::198 или 2001::199
- DNS: 2001::1
- Пинг до 2001::1 успешен

PC3 IPv6 полный

Рис. 17: PC3 IPv6 полный

Просмотр аренд DHCPv6 Stateful

```
[edit]
zashikhalieva@zashikhalieva-gw-01# run show dhcpv6 server leases
IPv6 address    State    Last communication    Lease expiration    Remaining
Type    Pool    IAID_DUID
--------------  -------  --------------------  ------------------  -----------
------  ------  -----------
[edit]
zashikhalieva@zashikhalieva-gw-01#
[edit]
zashikhalieva@zashikhalieva-gw-01# run show dhcpv6 server leases
```

Рис. 18: Stateful leases

Пакеты Solicit, Advertise, Request, Reply

No.	Time	Source	Destination	Protocol	Lengtl	Info
1	0.000000	fe80::42:b2ff:fe11:…	ff02::1:2	DHCPv6	116	Solicit XID: 0xe1a426 [ROOT-ONLY DOMAIN NAME]CID: 0003000107
2	0.005769	fe80::e52:26ff:feb8…	fe80::42:b2ff:fe11:…	DHCPv6	185	Advertise XID: 0xe1a426 IAA: 2001::199 CID: 000300010242b2114
3	0.005929	fe80::42:b2ff:fe11:…	fe80::e52:26ff:feb8…	ICMPv6	233	Destination Unreachable (Port unreachable)
4	0.026070	fe80::42:b2ff:fe11:…	ff02::1:2	DHCPv6	134	Request XID: 0xe1a426 [ROOT-ONLY DOMAIN NAME]CID: 0003000107
5	0.029219	fe80::e52:26ff:feb8…	fe80::42:b2ff:fe11:…	DHCPv6	185	Reply XID: 0xe1a426 IAA: 2001::199 CID: 000300010242b2114f00
6	0.029420	fe80::42:b2ff:fe11:…	fe80::e52:26ff:feb8…	ICMPv6	233	Destination Unreachable (Port unreachable)
7	5.214532	fe80::42:b2ff:fe11:…	fe80::e52:26ff:feb8…	ICMPv6	86	Neighbor Solicitation for fe80::e52:26ff:feb8:2 from 02:42:b2
8	5.218075	fe80::e52:26ff:feb8…	fe80::42:b2ff:fe11:…	ICMPv6	78	Neighbor Advertisement fe80::e52:26ff:feb8:2 (rtr, sol)
9	5.408496	fe80::e52:26ff:feb8…	fe80::42:b2ff:fe11:…	ICMPv6	86	Neighbor Solicitation for fe80::42:b2ff:fe11:4f00 from 0c:52:
10	5.409364	fe80::42:b2ff:fe11:…	fe80::e52:26ff:feb8…	ICMPv6	78	Neighbor Advertisement fe80::42:b2ff:fe11:4f00 (sol)
11	24.269167	fe80::42:b2ff:fe11:…	2001::1	ICMPv6	118	Echo (ping) request id=0x013a, seq=0, hop limit=64 (reply in
12	24.270588	2001::1	fe80::42:b2ff:fe11:…	ICMPv6	118	Echo (ping) reply id=0x013a, seq=0, hop limit=64 (request in
13	25.269199	fe80::42:b2ff:fe11:…	2001::1	ICMPv6	118	Echo (ping) request id=0x013a, seq=1, hop limit=64 (reply in
14	25.272076	2001::1	fe80::42:b2ff:fe11:…	ICMPv6	118	Echo (ping) reply id=0x013a, seq=1, hop limit=64 (request in

```
> Frame 1: Packet, 116 bytes on wire (928 bits), 116 bytes captured (928 bits) on interface -, id 0
> Ethernet II, Src: 02:42:b2:11:4f:00 (02:42:b2:11:4f:00), Dst: IPv6mcast_01:00:02 (33:33:00:01:00:02)
> Internet Protocol Version 6, Src: fe80::42:b2ff:fe11:4f00, Dst: ff02::1:2
> User Datagram Protocol, Src Port: 546, Dst Port: 547
v DHCPv6
    Message type: Solicit (1)
    Transaction ID: 0xe1a426
  > Elapsed time
  > Identity Association for Non-temporary Address
  > Option Request
  > Client Fully Qualified Domain Name
  > Client Identifier
```

Итоги

Основные достижения

- Настроен DHCP для IPv4
- Настроены SLAAC, DHCPv6 Stateless и Stateful
- Выполнена полная проверка маршрутизации и связности
- Проанализирован сетевой трафик всех DHCP-механизмов
- Подтверждена корректная работа всех служб распределения адресов